

Backpack App Requirements Document

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1. INTRODUCTION

McCann wishes to better understand the cost of a completed job, the team believes an application enabling operatives to fill in job details would assist with this, as well as enabling the intended job cost calculation to be performed by the QS

This document presents the requirements of the Backpack application specification that will be satisfy, from a technical and system operations perspective in relation to the business process it will support.

The Backpack app is developed by the KTP Associate in collaboration with the Business Partner Supervisor at McCann and the Knowledge Base Supervisor, University of Suffolk.

1.1. SCOPE

This requirements specification applies to the task version completed as part of the KTP recruitment process, it is built to demonstrate my existing experience in Software engineering.

2. ROLES & RESPONSIBILITIES

Role	Responsible for
Operative	- Record and update site visit record - Record materials used during site visits
QS	- Review and approve visit records - Manage rates for operatives and materials

3. DEFINITIONS

Brief business and technical definitions specific to the business process that may be used in this document in relation to the system are given below:

Operative	Worker on site who completes hands on work within a location, e.g. a "job" such as repairing a streetlight
Quantity Surveyor (QS)	An Office-based staff who completes the financial parts of each job
Site visit	A session of time where an operative completes hands on work within a location

4. OVERVIEW OF THE SYSTEM

4.1. INTENDED USE OF THE SYSTEM

The Backpack app will be used by Operatives and Qs to report and manage calculations of completed jobs respectively.

Records of site visits will be recorded by the Operative, including any resources used, the cost of these items are calculated as well as cost of working operatives on the job, this is presented for the QS to complete the financial part of the job. The system will also allow for the Qs to create jobs with planned costs to compare against actuals.

The key purpose of the system is for McCann and her teams to understand better the cost of completed jobs.

4.2. DESCRIPTION OF BUSINESS PROCESS

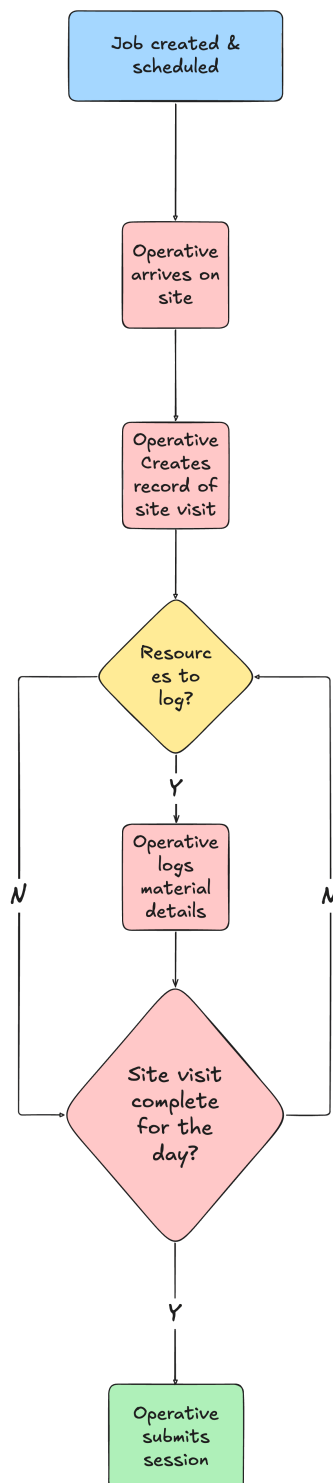


Figure Site visit recording

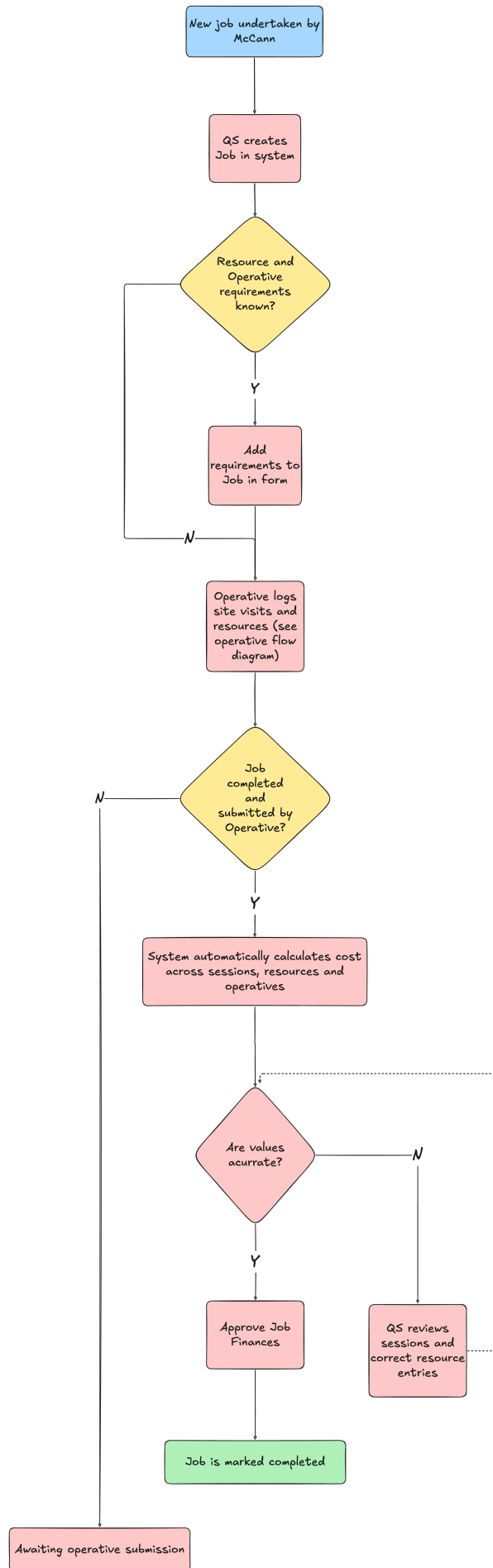


Figure QS cost management

5. REQUIREMENTS

Reference	Requirement
R-MCANN-AUTH-1	Users must authenticate with an email address and password
R-MCANN-AUTH-2	Role-based access control is enforced at the middleware level; such that Operatives can't access QS routes and vice versa
R-MCANN-JOB-1	QS users can create jobs with a reference number, name, site, start date and expected headcount
R-MCANN-JOB-2	Jobs have a status lifecycle: Draft → Active → Completed → Approved
R-MCANN-JOB-3	QS users can update job status and mark jobs as complete
R-MCANN-JOB-4	Each job is associated with a site and tracks its sessions, operatives and total cost
R-MCANN-SESSION-1	Operatives create a session to record a site visit, capturing job, start time and optional notes
R-MCANN-SESSION-2	An operative may only have one open (unsubmitted) session at a time, this is enforced at both application and database level, reflecting that an operative can only be physically present at one site simultaneously
R-MCANN-SESSION-3	A session is submitted by the operative when the site visit is complete for the day, recording the end time and submitting operative
R-MCANN-SESSION-4	Submitted sessions cannot be modified
R-MCANN-RESOURCE-1	Operatives log resources used within a session across three categories; Materials, Tools, and Mechanical
R-MCANN-RESOURCE-2	The rate applicable to each resource is snapshotted at the time of recording to preserve historical accuracy regardless of future rate changes
R-MCANN-RESOURCE-3	Calculated cost is stored alongside the rate snapshot
R-MCANN-COST-1	The system automatically calculates costs across all sessions, operatives and resources when a job is marked complete
R-MCANN-COST-2	Operative costs are derived from the snapshotted rate and the duration between arrival and departure times

6. DEVOPS INTEGRATION

To integrate DevOps practices the project will be developed using a Version Control System (VCS) such as Git, which will allow shared ownership between team members developing the project, this approach also allows collaboration such that the team can iterate on several parts of the Backpack App separately and deliver features at speed.

The project will also integrate Automation which allows for performing repeated tasks relating to the delivery of the project such as deploying the app to the end once features have been approved and completed, and integrating risk reduction practices such as testing, dependency scanning etc.

In this project we will use a platform such as Github, which also uses a distributed version control system built on Git as well as CI/CD platform to allow automations.

7. APPENDICES

7.1. APPLICATION ARCHITECTURE ARCHITECTURE

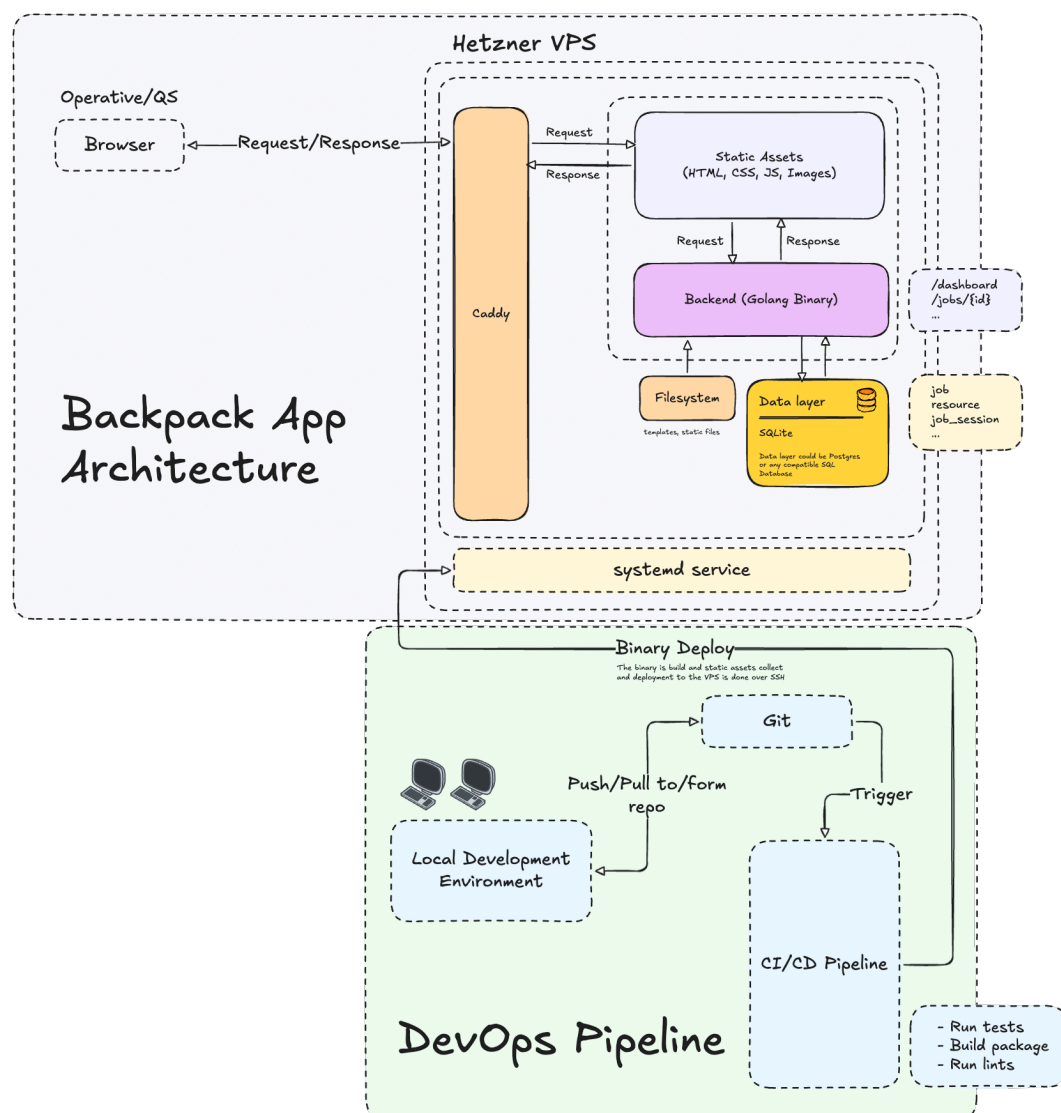


Figure Backpack App Architecture with DevOps Pipeline